

Pill Buddy by Abhiyantrix

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Motivation

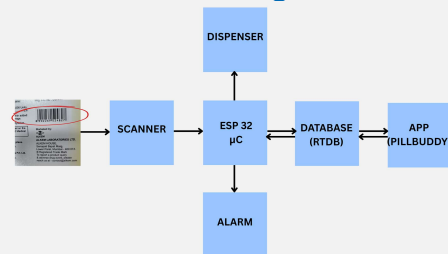
- **Medication non-adherence** is a major issue in chronic disease management.
- **Over-the-counter (OTC) drug usage is not tracked** in home settings.
- Misuse of antibiotics contributes to increasing **antibiotic resistance**.
- Existing pill dispensers mainly provide basic reminders without monitoring adherence.
- Lack of integrated systems that **combine dispensing, monitoring, and data recording (medicine history tracking)**.

There is a current need for a smart medication dispensing system that ensures timely drug intake and tracks patient medication history.

State of the art

- Automated dispenser holds 90-day supply, app alerts, one-button dispense (Hero Health)
- Cloud dashboard for caregivers, real-time alerts, reporting. (MedMinder)
- Automatic dispensers with reminders, locking, family app sharing (LiveFine)
- Smart bottles with weight sensors, real-time tracking (AdhereTech)
- Smart dispenser with cap sensors, MEMS software for adherence analytics (Pill Connect)

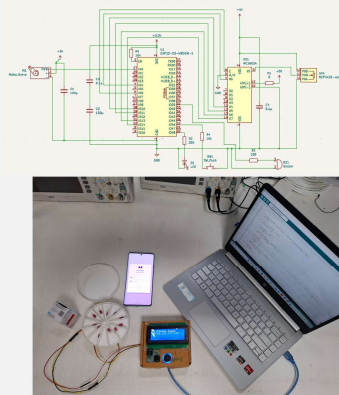
Core technological idea



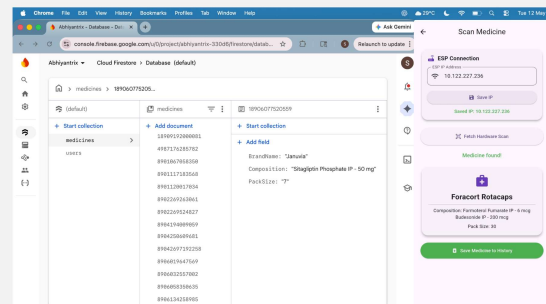
Implementation & Achievements



Pill Dispenser (Hardware)



Digital Interface and database



Reflection & Conclusion

- Developed an IoT-based smart pill dispenser using ESP32, Flutter, and Firebase RTDB.
- Built a local 15-medication database to avoid third-party scraping limitations.
- Designed a 3D dispensing mechanism and resolved singulation and universality issues.
- Integrated barcode scanning, alarms, LCD, buzzer, LEDs, and servo-based dispensing.
- Achieved real-time Wi-Fi communication between the app, cloud database, and hardware.
- Migrated from breadboard to finalized PCB, resulting in a fully functional prototype.

Outlook

- GSM integration for reduced Wi-Fi dependency.
- AI-based pattern recognition and doctor alert system.
- ABDM integration for medication tracking and monitoring.
- Increased storage capacity with advanced dispensing mechanisms.

References

- 1 <https://www.techenhancedlife.com/reviews/hero-medication-dispenser>
- 2 <https://medminder.com/>
- 3 <https://www.livefineproduct.com/collections/pill-dispensers>
- 4 <https://adheretech.com/>
- 5 <https://www.pillconnect.com/>